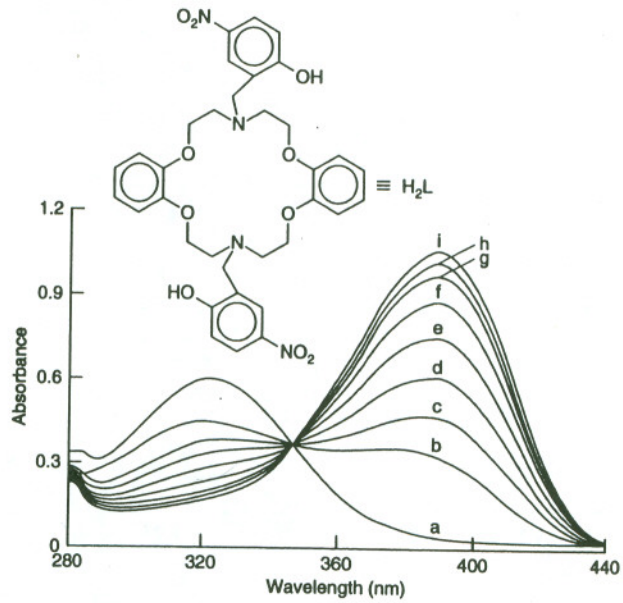
Analytical Cumulative Examination
October 8, 1999

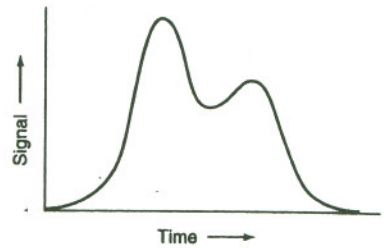
Separation Science

1. The figure at right shows spectral changes for a recently developed, crown ether extraction reagent, obtained in an organic solvent before (a) and after (b-i) extracting Hg (II) from aqueous solution as a function of pH: (b) pH=2.0; (c), pH=2.2; (d) pH=2.5; (e) pH=2.7; (f) pH=3.0; g) pH=3.2; (h) pH=3.4; and (I) pH=3.9. H_2L is soluble in the organic solvent, has negligible solubility in water, and the metal-reagent complex has no net charge.



   a. Draw the structures of H_4L^{2+} and L^{2-}.
   b. Briefly explain the observed spectral changes in the organic phase. Which species is responsible for the maximum at 320 nm, and which is responsible for the maximum at 388 nm?
   c. Why does the 388 nm peak grow as the pH of the extracted solution increases.
   d. What does the isosbestic point at 348 nm show?
   e. Briefly explain how pH control might allow selective extraction; that is, the extraction of one metal ion in the presence of another, given that the formation constants for the two metal complexes are different.

2. Two peaks elute from a chromatography column as shown to the right: Resolution, (R) is given as

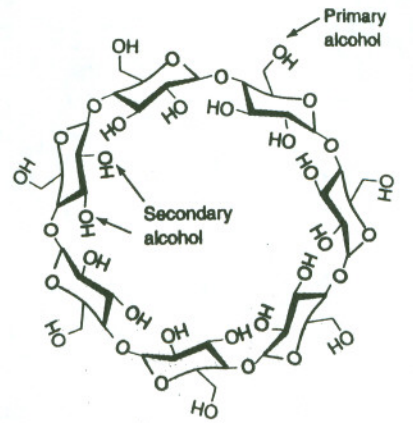


$$R = \frac{\sqrt{N}}{4}\left(\frac{\alpha - 1}{\alpha}\right)\left(\frac{k'_{avg}}{1 + k'_{avg}}\right)$$

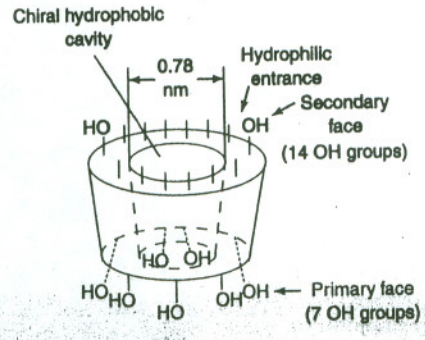
where N is the number of theoretical plates, α is the relative retention and k'_{avg} is the average of the capacity factors for the 2 peaks.

   a. Sketch what the chromatogram would look like if N and α were kept constant, but k'_{avg} was increased. Explicitly describe how would this be accomplished in Reverse Phase-HPLC without changing column length.
   b. Sketch what the chromatogram would look like if N and k'_{avg} were kept constant but α was increased. Explicitly describe how this would be accomplished in HPLC.
   c. Sketch what the chromatogram would look like if α and k'_{avg} were kept constant but N was increased. Explicitly describe how this could be accomplished in HPLC.

3. To the right is the structure of β-cyclodextrin, a cyclic polysaccharide composed of 7 sugar molecules. The cavity dimension is well defined, the interior is hydrophobic, while the exterior is hydrophillic. Briefly describe how one could use this reagent to separate enantiomers in HPLC. What factors would determine whether a separation would be possible with this reagent.



(a)



(b)

4. Capillary electrophoresis (CE) involves the migration of ions in solution using a fused silica capillary, under the influence of an electric field.

   a. Briefly describe the basis for electro-osmotic flow (EOF), noting that the pK_a for Si-OH is ~2, and EOF is observed at pH's>2, with the velocity increasing with pH.
   b. Describe (diagram) the flow profile for EOF versus typical pressure driven flow characteristic of GC or HPLC.
   c. Qualitatively compare the maximum separation efficiency for capillary column GC (nonpacked), HPLC and CE. To do this, consider the Van Deemter equation and the three main contributions to plate height (height equivalent to theoretical plate): longitudinal diffusion, multiple paths and resistance to mass transfer.